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MILITARY SPECIFICATION

PINS, TENT, WOOD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and three sizes of wood tent pins.

1.2 Classification. Wood tent pins shall be of the following sizes as specified (see 6.2).

Size 1 - 16 inch
Size 2 - 24 inch
Size 3 - 36 inch

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel And Seals
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-4-1 - Pins, Tent, Wood
5-4-414 - Pins, Tent, Wood; Containers for Domestic and Overseas Shipment

* (Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of docu-ments not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 512 - Cold-Drawn Buttweld Carbon Steel Mechanical Tubing
A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing
D 2016 - Moisture Content of Wood
* D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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* 2.3 order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials (see 6.4). The materials used shall conform to the applicable referenced documents specified herein. Where materials are not definitely specified, they shall be of the quality and type normally used by the manufacturer provided the complete item complies with all the provision of this document.

3.2.1 Wood. Wood shall be uniformly dried without brashness, honey-combing or casehardening to a moisture content of not less than 8 nor more than 18 percent at time of fabrication when tested as specified in 4.4.1. Wood shall weigh a minimum of 36 pounds per cubic foot at 12 percent moisture content when tested as specified in 4.4.1 by any commercial testing method. The species of wood used shall be any of the following:

Ash, white	Hickory (true or pecan), Kerving/Apitone
Beech	Locust, black or honey
Birch, sweet or yellow	Maple, sugar or black (hard)
Elm, rock	Oak, red or white, Phillipine hardwood or Malaysian hardwood

3.2.2 Tube, steel. Steel tube shall conform to MT 1010- 1020, any type or condition of ASTM A 512 or A 519.

3.2.3 Screw, wood. Wood screw shall be of commercial quality zinc plated steel and of the type and size shown on Drawing 5-4-1.

3.3 Construction. Construction of tent pins shall conform to Drawing 5-4-1 and as specified herein.

3.3.1 Pins. Pins shall be fabricated of wood specified in 3.2.1. Pins shall be cut in the direction of the grain of the wood as shown on Drawing 5-4-1. After cutting the pins to shape, the slope of the wood grain shall not exceed 1 inch in 10 inches when tested as specified in 4.4.2.1. The surfaces on all sides of the pins shall be free from torn grain having a depth greater

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than 1/16 inch and skips having an area greater than 1 square inch. Any hole having diameter greater than 1/8 inch will be acceptable provided the hole is completely filled with an epoxy filler, the epoxy filling within the hole is sound and tight, and complies with other requirements herein. Knots shall be sound and tight. Irregular local grain deviations such as birdseye, burl or swirl will be permitted. All pins shall be free from the following:

a. Knots or epoxy filled holes less than 3 inches apart
b. Heavy stain completely obscuring the grain of the wood
c. Wormholes having diameter greater than 1/16 inch or wormholes closely bunched together exceeding more than 5 in an area of 1/2 square inch on any surface.
d. Holes (other than wormholes) having diameter greater than 1/8 inch or spaced closer than 2 inches apart
e. Splits, loosened grain, larvae channel, bark or gum pocket
f. Decay

3.3.1.1 Sizes 1 and 2. Size 1 pin shall be free from any knot or epoxy filled hole within 6 inches from the top or within 3 inches from the bottom. Size 2 pin shall be free from any knot or epoxy filled hole within 11 inches from the top or within 3 inches from the bottom. Sizes 1 and 2 pins shall be free from the following:

a. Knot or epoxy filled hole having diameter greater than 1/4 inch on the 1-1/16 inch face or diameter greater than 3/8 inch on the 1-1/2 inch face
b. More than 2 knots or epoxy filled holes per pin (applicable to size 1 pin)
c. More than 4 knots or epoxy filled holes per pin (applicable to size 2 pin)
d. Surface check open more than 1/64 inch wide or more than 1-1/2 inches long
e. More than a total of 8 surface checks on the 4 surfaces
f. Surface checks closer than 1/4 inch from any edge
g. More than 4 holes (other than wormholes) on a face (applicable to size 1 pin)
h. More than 6 holes (other than wormholes) on a face (applicable to size 2 pin)

3.3.1.2 Size 3. The size 3 pin shall be fitted with a ferrule specified in 3.3.2. The ferrule shall be secured on the pin with a screw specified in 3.2.3. Size 3 pin shall be free from the following:

a. Knot or epoxy filled hole on any face having diameter greater than 1/2 inch
b. Knot or epoxy filled hole within 3 inches for the top or within 12 inches from the bottom
c. More than 6 knots or epoxy filled holes per pin
d. Surface check more than 1/64 inch wide or more than 3 inches long on any surface of pin

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e. More than 16 surface checks on the whole pin
f. More than 8 holes (other than wormholes) on a single face

3.3.1.2.1 Driving of screw. Screw shall be screwed in place with head flush with the ferrule. Screw shall not be driven with a hammer.

3.3.2 Ferrule. Ferrule shall be fabricated of steel specified in 3.2.2. The hole in the ferrule shall be countersunk to provide a flush fit for the screw head.

3.4 Finish. Rough or sharp edges, burrs or slivers on ferrule shall be removed prior to finishing. Ferrule shall then be galvanized in accordance with commercial practice to produce an adherent continuous uniform coating free from scratch, dig, dirt or foreign matter imbedded in coating.

3.5 Workmanship. The finished pins shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table III and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Component testing. In addition to the quality assurance provisions of the subsidiary documents, testing shall be performed on materials listed in table I for the test characteristics shown. Requirements for moisture content and weight of wood are applicable to the individual unit. The acceptable quality level (AQL) shall be expressed in terms of percent defective. Any sample unit of wood tested having less than 8 or more than 18 percent moisture content shall be classified as defective. Any sample of wood tested weighing less than 36 pounds per cubic foot at 12 percent moisture content shall be classified as defective.

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TABLE I.

Materials and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	determinations per sample unit	Sample unit	Results reported as	Inspection level	AQL
Wood:								
* (100 board foot)	Moisture content	3.2.1	ASTM D 2016, Method A or B	3	1 representative piece of lumber	Average of 3 determinations to nearest 0.5 percent	S-1	6.5
	Weight	3.2.1	Commercial	1	1 representative piece of lumber	To nearest 0.5 pound per cubic foot	S-1	6.5

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4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether the required operations are accomplished as specified. Process inspection shall be conducted to see that accomplishment of the following is in accordance with the specified requirements. When nonconformance is noted, correction shall be made to affected items and lot in process:

Requirement or operation	Requirement reference
Driving of screw (not with hammer)	3.3.1.2.1
Removal of rough or sharp edges, burrs or slivers prior to finishing	3.4

4.4.2.1 In-process testing. Testing shall be performed in accordance with table II for the characteristic shown therein. The lot shall be all the pins of each size offered for testing at one time. The sample units for the slope of the grain test shall be one (1) size 1 pin and one (1) size 2 pin after all machining has been performed. The sample unit for the size 3 pin shall be one (1) square or rectangular blank. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE II. In-process tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to	No. of determinations per sample unit	Results reported as
Wood:					
Slope of grain	3.3.1	4.5.1	Individual unit	1	Pass or fail

* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of pins. The sample unit shall be one completely fabricated pin. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total (major and minor combined) defects. When a knot, epoxy filled hole, or hole is observed during examination for defects, the diameter of the knot, epoxy filled hole, or hole shall be measured between lines parallel to the edge of the pin.

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TABLE III. End item defects

Examine	Defect	Classification Major	Minor
Finish (size 3 only):			
(applicable to ferrule)	Rough or sharp edge, burr, or sliver		X
	Not galvanized coating	X	
	Not adherent continuous uniform coating		X
	Scratch, dig, dirt or foreign matter imbedded in coating		X
(applicable to screw)	Not zinc plated		X
Construction (applicable to all sizes of pins)	Torn grain having a depth greater than 1/16 inch		X
	Skips having an area greater than 1 square inch		X
	Sharp longitudinal corners not removed		X
	Epoxy filled hole not completely filled		X
	Epoxy filling within the hole is not sound or not tight		X
	Knot not sound or not tight		X
	Sound knots or epoxy filled holes are less than 3 inches apart		X
	Heavy stain completely obscuring the grain of the wood		X
	Wormhole having diameter greater than 1/16 inch or wormhole closely bunched together exceeding more than 5 in an area of 1/2 square inch on any surface		X
	Holes (other than wormholes) having diameter greater than 1/8 inch or spaced closer than 2 inches apart	X	
	Split, loosened grain, larvae channel, bark or gum pocket	X	
	Decay	X	
(applicable to size 1 or 2 pins)	Sound knot or epoxy filled hole within 6 inches from top or within 3 inches from bottom (applicable to size 1 pin)		X

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TABLE III. End item defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction (applicable to size 1 or 2 pins) (cont'd)	Sound knot or epoxy filled hole within 11 inches from top or within 3 inches from bottom (applicable to size 2 pin)		X
	Sound knot or epoxy filled hole having diameter greater than 1/4 inch on the 1-1/16 inch face or diameter greater than 3/8 inch on the 1-1/2 inch face	X	
	More than 2 sound knots or 2 epoxy filled holes per pin (applicable to size 1 pin)		X
	More than 4 sound knots or 4 epoxy filled holes per pin (applicable to size 2 pin)		X
	Surface check open more than 1/64 inch wide or more than 1-1/2 inches long		X
	More than a total of 8 surface checks on the 4 surfaces		X
	Surface checks closer than 1/4 inch from any edge		X
	More than 4 holes (other than wormholes) on a face (applicable to size 1 pin)		X
	More than 6 holes (other than wormholes) on a face (applicable to size 2 pin)		X
(applicable to size 3 pin)	Not fitted with ferrule	X	
	Ferrule not secured on pin with screw		X
	Sound knot or epoxy filled hole on any face having diameter greater than 1/2 inch	X	
	Sound knot or epoxy filled hole within 3 inches from top or within 12 inches from bottom	X	
	More than 6 sound knots or 6 epoxy filled holes per pin		X

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TABLE III. End item defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction (applicable to size 3 pin) (cont'd)	Surface check more than 1/64 inch wide or more than 3 inches long on any surface of pin		X
	More than 16 surface checks on the whole pin		X
	More than 8 holes (other than wormholes) on a single face		X
	Screw hole in ferrule not countersunk		X
	Screw head not flush fit with ferrule		X

* 4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on Drawing 5-4-1. Any dimension not within specified tolerances shall be classified as a defect. the lot size shall be expressed in units of pins. The sample unit shall be one completely fabricated pin. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

* 4.4.5 Packaging examination. The fully packaged end item shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method or application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: loose strapping Bulged or distorted container Arrangement of pins within containers not as specified on Drawing 5-4-414

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Examine	Defect
Content	Quantity of pins per container is more or less than required More than one size of pin in any container

* 4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limit.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Slope of wood grain. The slope of grain may be determined by combining the grain direction of two adjacent faces of a square or rectangular piece. When one of these faces is straight grained, the true slope of grain is shown on the other face. When there is a slope on both faces, the true or combined slope is greater than the slope on either face. Slope of grain is shown on edge grained faces by the summer wood bands, by the direction in which a free flowing ink or dye spreads, or by the course taken by a narrow strip lifted by a knife point and torn out. Direction of fiber may also be measured by a slope of grain detector. Any size 1 pin, size 2 pin, or square or rectangular blank of the size 3 pin (see 4.4.2.1) having a slope of wood grain exceeding 1-inch in 10 inches shall be classified as defective.

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5. PACKAGING

* 5.1 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A packing. Tent pins shall be arranged and packed in containers as shown in Drawing 5-4-414. Quantities per container shall be as follows:

Size 1 pins - 150 per container
Size 2 pins - 100 per container
Size 3 pins - 15 per container

Containers shall be constructed to conform to PPP-B-621, except that the top and bottom shall be of slatted construction as shown on Drawing 5-4-414. Each container shall be strapped with steel straps, 5/8 by 0.020 inch conforming to type I, regular duty, finish B, grade 2 of QQ-S-781. Alternatively, round galvanized strapping 0.1055 inch in diameter conforming to class 2, type V, finish B, of QQ-S-781 may be used.

* 5.1.2 Level B packing. Tent pins shall be arranged as shown on Drawing 5-4-414 with quantities per container as specified in 5.1.1. Containers shall conform to style FTC, type CF (variety SW) or SF, class domestic, grade 350 of PPP-B-636. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. Inside dimensions of the shipping container shall be as shown in Drawing 5-4-414.

* 5.1.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

* 5.1.3 Commercial packing. Tent pins shall be packed in accordance with ASTM D 3951.

* 5.2 Palletization. When specified (see 6.2), tent pins, packed as specified in 5.1, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet types shall be type 1 (4-way entry), type IV or type V, in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 for container sizes 1 and 2, and pallet pattern number 2 for container size 3. Interlocking of loads shall be effected by reversing the pattern of each course.

* 5.3 Marking. In addition to any special marking required by the contract, or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

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6. NOTES

6.1 Intended use. The tent pins are used with various tents, tent flies, and latrine screens.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Size of tent pin required (see 1.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. Selection of applicable level of packing (see 5.1).
e. When weather-resistant fiberboard containers are required for level B packing (see 5.1.2.1).
f. When palletization is required (see 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52-209. The first article should be a preproduction sample, consisting of one unit of each size of tent of pin required. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2882, relative to camouflage requirements for tents, shelters and subsidiary components. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

* 6.6 Subject term (key word) listing.

Pins
Shelters
Tents

6.7 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:

Air Force - 82
DLA - CT

User activities:

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Air Force - 45

Preparing activity:

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